

Engg. Mechanics.

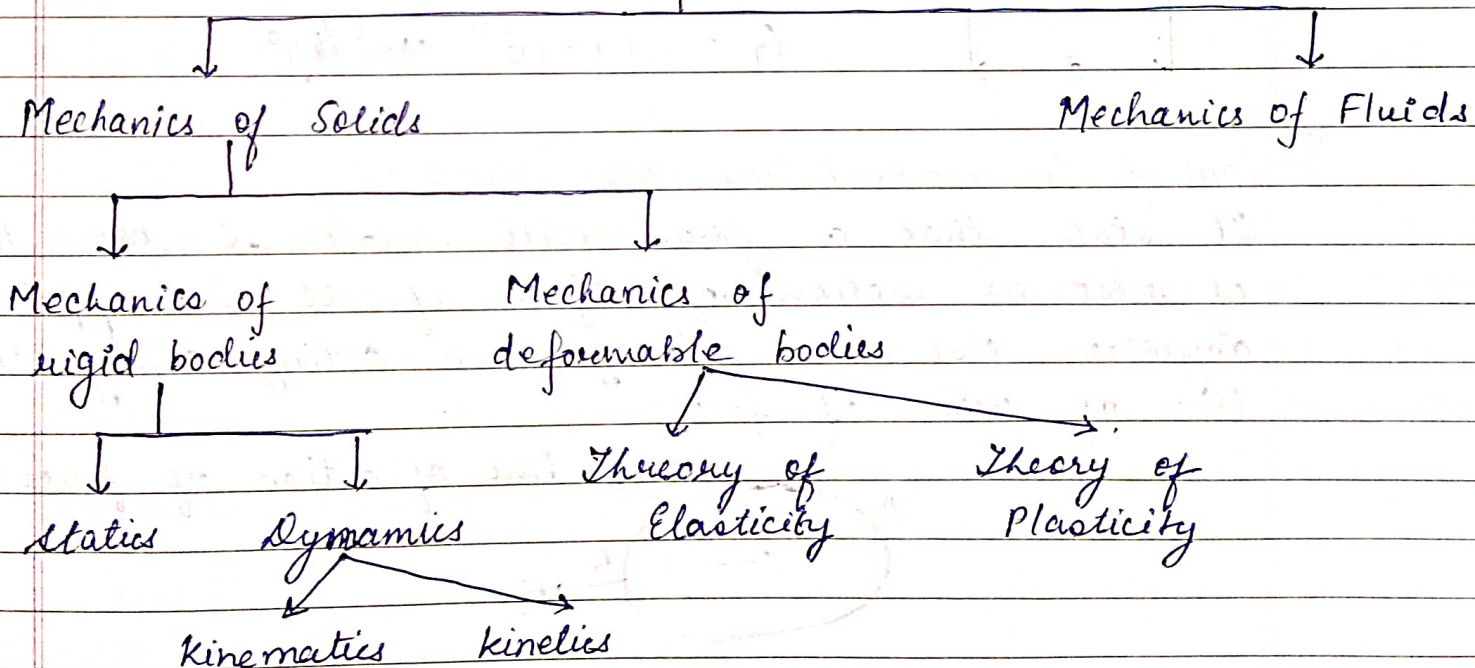
- It is a branch of physical science which deals with the object at rest or in motion under the action of a force.
- Application of laws of mechanics to field problem is known as Engg. Mechanics.
- ① classical Mechanics or Newtonian Mechanics.
- ② Relativistic Mechanics
- ③ Quantum Mechanics.

Classical:- Based upon laws of motion and rigid bodies

Relativistic:- Study the motion of object at speed of light

Quantum:- Study the motion of atomic, molecular level

Engg. Mechanics

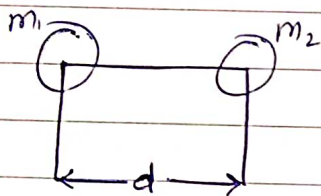


Basic Terminology:-

- ① Mass
- ② Time
- ③ Length:- It is a concept of measuring linear distance.
- ④ Displacement:- Relative position
- ⑤ Velocity
- ⑥ Acceleration
- ⑦ Momentum
- ⑧ Continuum

Laws of Mechanics:-

- ④ Laws of Gravitation
- ① Newton's 1st law of Motion
- ② Second law of Motion
- ③ Third law of Motion
- ⑤ Law of Transmissibility of force:-

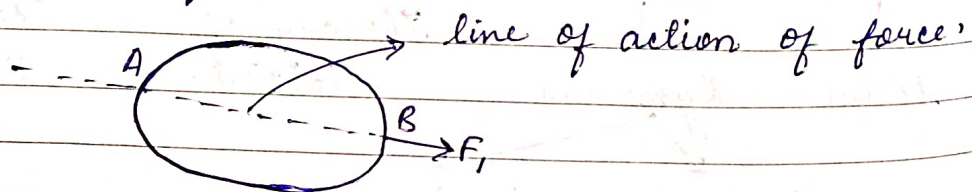


$$F = \frac{G m_1 m_2}{d^2}$$

$$G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$$

⑤ Law of Transmissibility of force:-

It states that a force will unaltered the state of rest or motion of object if it is applied anywhere else on a body but acting along the line of action of force.



⑥ Parallelogram ^{law} of force:-

This law states that if two forces acting simultaneously on a body at a point are represented in

Force:- force is the agency which causes the state of a body to change either it is in rest or motion.

S.I unit of force $\rightarrow$ Newton.

1N is defined as force required to move 1kg mass with an acceleration of 1 m/s^2

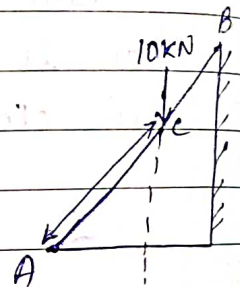
$$1\text{N} = 1 \text{ kg m/s}^2$$

$$G = M^{-1} L^3 T^{-2}$$

Characteristics of Force

It is completely specified when following four characteristics are given:-

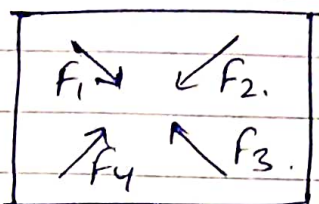
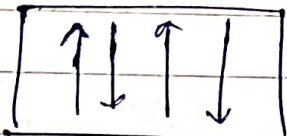
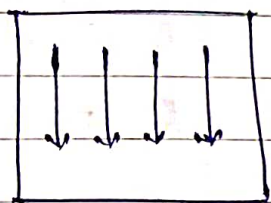
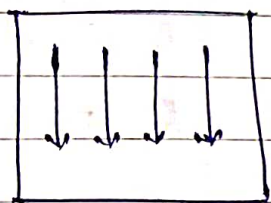
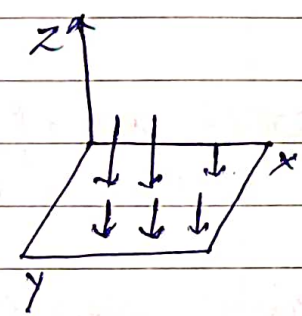
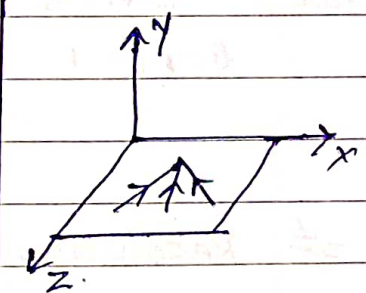
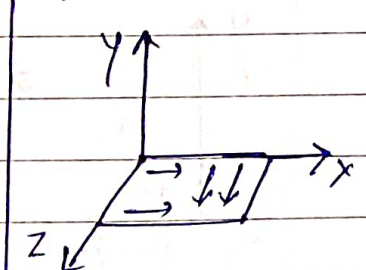
- Magnitude
- Direction
- Line of action
- Point of application



System of Forces

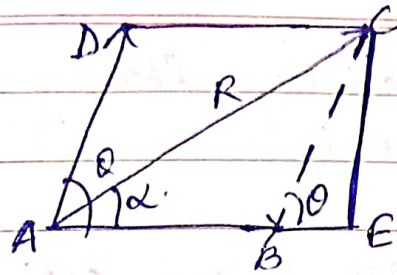
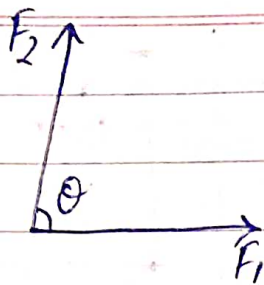
When several forces are acting on a body simultaneously called as System of Forces.

System of force	Characteristics	Fig.
Collinear force	All the forces acting on the same line of action	
Coplanar concurrent force	All the forces must act on a single plane and must pass through single point	
Coplanar non-concurrent force	All forces lie on a single plane but not	

	passing through single point	 
Coplanar parallel forces	All forces acting on a single plane and parallel to each other. e.g. forces acting on beam	
Coplanar like parallel forces.	All forces lie on a single plane and acting in same direction	
Non-coplanar and parallel forces.	All forces will not act in a single plane but act in a same direction.	
Non-coplanar concurrent forces.	All forces are acting in different planes but pass through a single point.	
Non-coplanar non-concurrent force.	Forces are acting in different planes and different directions.	

Resultant and Equilibrium of coplanar and concurrent forces.

Composition of Two force System



Drop $\perp$ to AB.
from C

$$AC = \sqrt{(AE)^2 + (CE)^2}$$

$$BE = F_2 \cos \theta$$

$$R = \sqrt{(AB + BE)^2 + (CE)^2}$$

$$CE = F_2 \sin \theta$$

$$R = \sqrt{(F_1 + F_2 \cos \theta)^2 + (F_2 \sin \theta)^2}$$

$$R = \sqrt{F_1^2 + F_2^2 \cos^2 \theta + 2F_1 F_2 \cos \theta + F_2^2 \sin^2 \theta}$$

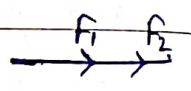
$$R = \sqrt{F_1^2 + F_2^2 + 2F_1 F_2 \cos \theta}$$

Case 1: when $\theta = 90^\circ$

$$R = \sqrt{F_1^2 + F_2^2}$$

Case 2: When $\theta = 0^\circ$

$$R = F_1 + F_2$$

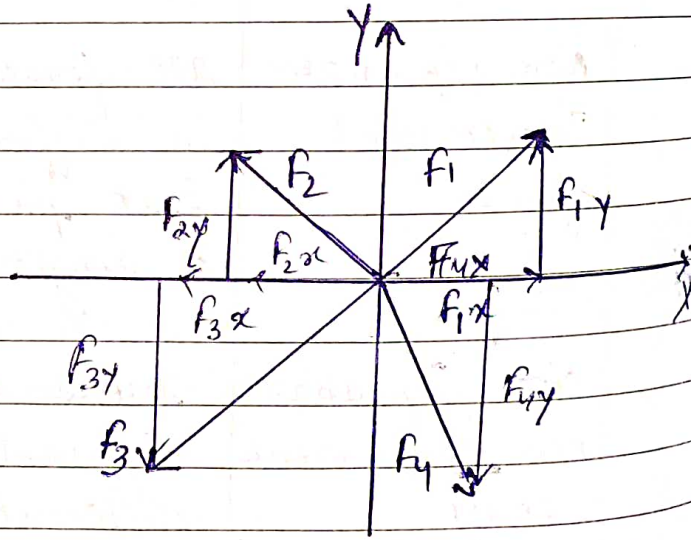
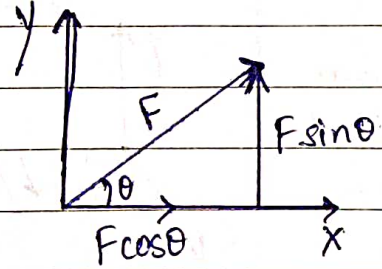


Case 3: When $\theta = 180^\circ$

$$R = F_1 + F_2$$



Resolution of Forces.

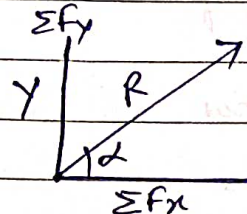


General composition
of coplanar-concurrent
forces.

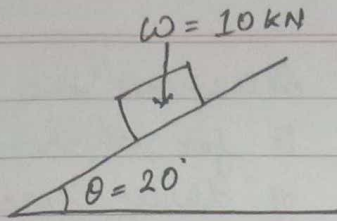
$$\Sigma F_x = F_{1x} + F_{2x} + F_{3x} + F_{4x}$$

$$\Sigma F_y = F_{1y} + F_{2y} + F_{3y} + F_{4y}$$

$$\tan \alpha = \frac{\Sigma F_y}{\Sigma F_x}$$



Ques



Find the rectangular components of weight along plane and $\perp$ to plane.

$$R = \sqrt{F_x^2 + F_y^2}$$

$$\Sigma F_x = R \cos \alpha$$

$$\Sigma F_y = R \sin \alpha$$

Equilibrium of bodies :- net force acting on body zero.

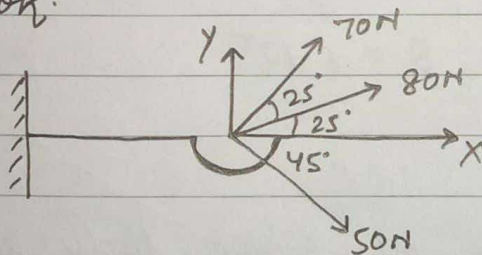
Equilibrium condition :- $R=0$ $\Sigma F_x=0$ $\Sigma F_y=0$

Types of forces :-

- 1) Applied force
- 2) Non-applied force

Applied Force :- It is a force acts on a body externally or when a force is applied externally to a body is called applied force.

Ques Determine the resultant of three forces acting on a hook.

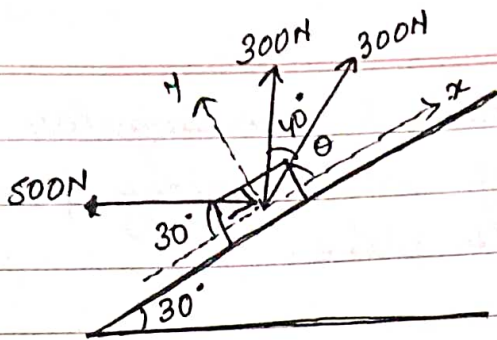


$$\begin{aligned} F_x &= 70 \cos 50^\circ + 80 \cos 25^\circ + 50 \cos 45^\circ \\ &= 44.99 + 72.50 + 35.35 \\ &= 152.84 \text{ N} \end{aligned}$$

$$\begin{aligned} F_y &= 70 \sin 50^\circ + 80 \sin 25^\circ - 50 \sin 45^\circ \\ &= 53.62 + 33.80 - 35.35 \\ &= 52.07 \text{ N} \end{aligned}$$

$$R = \sqrt{F_x^2 + F_y^2}$$

Ques



Determine the value of θ for which resultant of three forces is direction parallel to plane.

$$\begin{aligned}
 F_y &= -500 \sin 30^\circ + 300 \sin(40^\circ + \theta) + 300 \sin \theta \\
 0 &= -250 + 300 \{ \sin 40^\circ \cos \theta + \cos 40^\circ \sin \theta \} + 300 \sin \theta \\
 250 &= 300 \sin 40^\circ \cos \theta + 300 \cos 40^\circ \sin \theta + 300 \sin \theta \\
 &= 192.83 \cos \theta + 229.81 \sin \theta + 300 \sin \theta \\
 &\text{Complicated.}
 \end{aligned}$$

Another method

$$\begin{aligned}
 F_y &= 300 \sin \theta + 300 \sin(40^\circ + \theta) - 250 = 0 \\
 250 &= 300 \{ \sin \theta + \sin(40^\circ + \theta) \} \\
 5 \cdot 250 &= \frac{300}{6} \left\{ 2 \sin \left(\frac{40 + 2\theta}{2} \right) \cos 20^\circ \right\}
 \end{aligned}$$

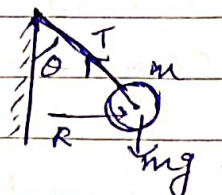
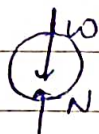
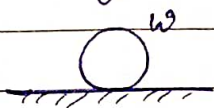
$$\frac{5}{12} = \sin(20^\circ + \theta) \cos 20^\circ$$

$$\frac{0.416}{\cos 20^\circ} = \sin(20^\circ + \theta) = \frac{0.416}{0.94} = 0.44$$

$$20^\circ + \theta = 26.10^\circ \Rightarrow \theta = 6.10^\circ$$

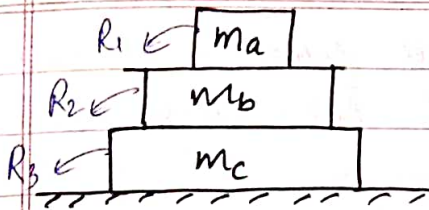
Free Body Diagram (FBD) :-

Such diagram of the body in which body under consideration is free from all the contact surfaces and is shown with all forces acting on it. (self weight and Reaction forces) is called free body diagram.

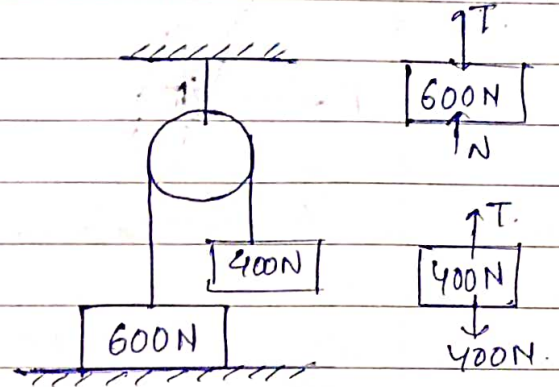
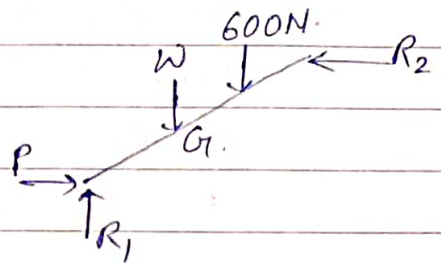
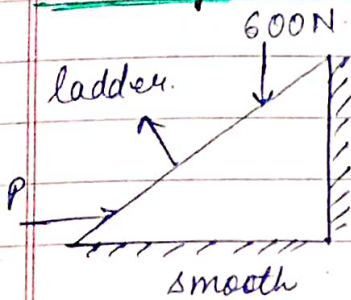


Ques

Draw the free body diagram of bodies

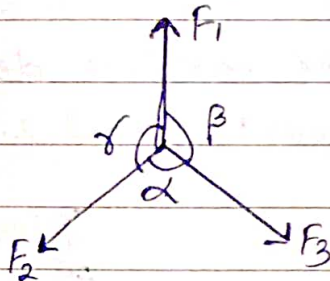


FBD of ladder.

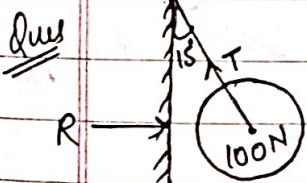


Lami's Theorem.

It states that when three forces are acting on a body in equilibrium then each force is proportional to sine of the angle between other two forces.

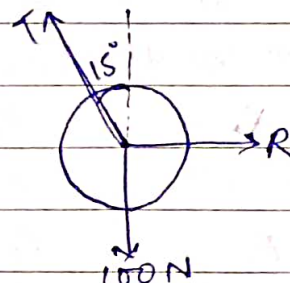


$$\frac{F_1}{\sin \alpha} = \frac{F_2}{\sin \beta} = \frac{F_3}{\sin \gamma}$$



Find Tension and Reaction.

Soln:-



Apply Lami's theorem

$$\frac{T}{\sin 90^\circ} = \frac{R}{\sin(180^\circ - 15^\circ)}$$

$$= \frac{100}{\sin(90^\circ + 15^\circ)}$$

$$T = \frac{R}{\sin 165^\circ} = \frac{100}{\sin 105^\circ}$$

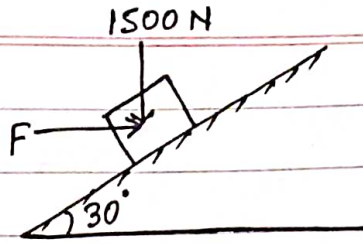
$$T = \frac{R}{0.258} = \frac{100}{0.965}$$

$$R = 0.258 \times T$$

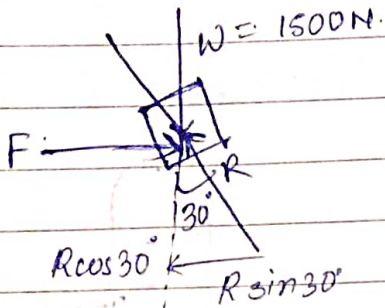
$$= 26.735 \text{ N.}$$

$$T = 103.62 \text{ N}$$

Ques



Determine force F required to hold the block in equilibrium.



$$\Sigma F_y = 0$$

$$R \cos 30^\circ - 1500 = 0 \quad \text{--- (1)}$$

$$\Sigma F_x = 0$$

$$F - R \sin 30^\circ = 0 \quad \text{--- (2)}$$

$$F = \frac{R}{2} \Rightarrow R = 2F$$

$$2F \cos 30^\circ = 1500$$

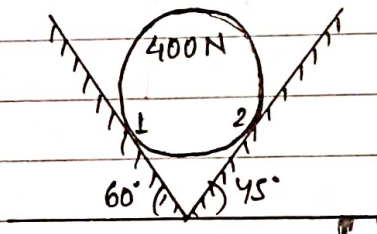
$$2F \times \frac{\sqrt{3}}{2} = 1500$$

$$F = \frac{1500}{\sqrt{3}} \text{ N} = 866 \text{ N}$$

Using Lami's theorem.

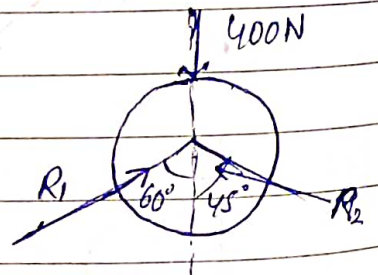
$$\frac{F}{\sin(180^\circ - 30^\circ)} = \frac{R}{\sin 90^\circ} = \frac{W}{\sin(90^\circ + 30^\circ)}$$

Ques



Determine the reactions developed at contact points 1 and 2.

Using Lami's theorem.



$$\frac{400}{\sin(60^\circ + 45^\circ)} = \frac{R_1}{\sin(180^\circ - 45^\circ)} = \frac{R_2}{\sin(180^\circ - 60^\circ)}$$

$$\frac{400}{\sin 105^\circ} = \frac{R_1}{\sin 135^\circ} = \frac{R_2}{\sin 120^\circ}$$

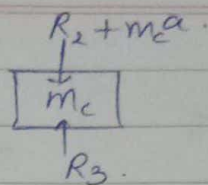
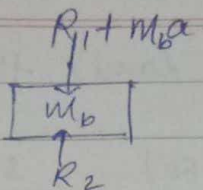
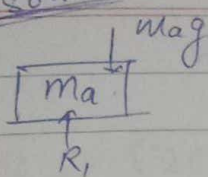
$$\sin 120^\circ = \frac{\sqrt{3}}{2}$$

$$\frac{400}{0.965} = \frac{R_1}{0.707} = \frac{R_2}{0.866}$$

$$R_1 = \frac{400 \times 0.707}{0.965} = 293 \text{ N}$$

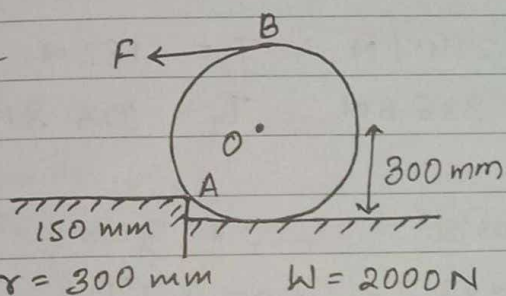
$$R_2 = \frac{400 \times 0.866}{0.965} = 358.6 \text{ N}$$

Homework Soln



If a body is in equilibrium under the action of three forces only, they must be concurrent force.

Ques



Find the magnitude of F required to start the roller move over the curb.

In $\triangle OAC$

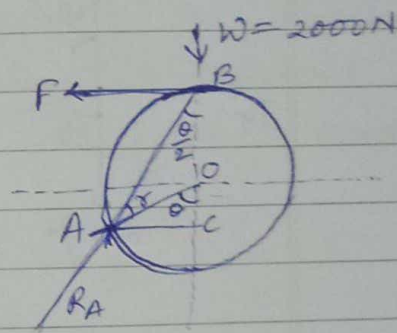
$$\cos \theta = \frac{OC}{OA} = \frac{150}{300} = \frac{1}{2} \quad \theta = 60^\circ$$

$$\text{So, } \angle OBA = 60/2 = 30^\circ$$

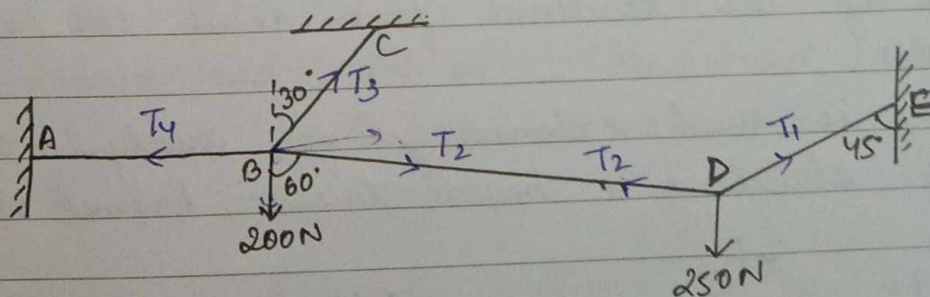
$$R_A \cos 30^\circ = 2000$$

$$R_A = \frac{2000 \times 2}{\sqrt{3}} = 2309.4 \text{ N}$$

$$R_A \sin 30^\circ = F = \frac{2309.4}{2} = 1154.70 \text{ N}$$

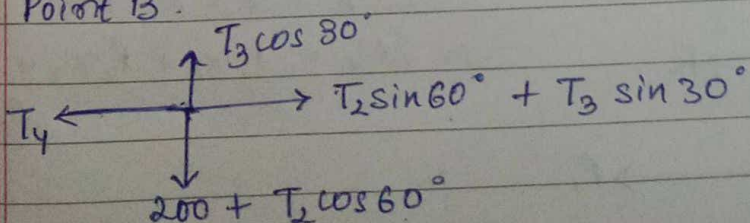


Equilibrium of Connected Bodies

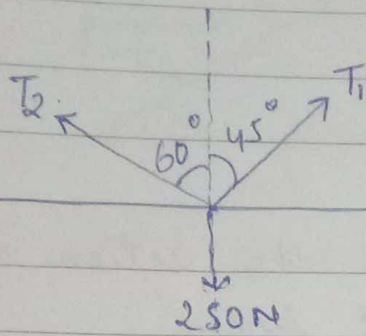


Ques Find the forces in various segments of cable.

Point B



FBD of D.



Using Lami's theorem.

$$\frac{T_1}{\sin(180^\circ - 60^\circ)} = \frac{T_2}{\sin(180^\circ - 45^\circ)} = \frac{250}{\sin(60^\circ + 45^\circ)}$$

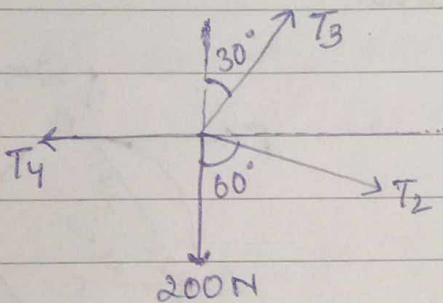
$$\frac{T_1}{\sin 120^\circ} = \frac{T_2}{\sin 135^\circ} = \frac{250}{\sin 105^\circ}$$

$$\frac{T_1}{0.866} = \frac{T_2}{0.707} = \frac{250}{0.965}$$

$$T_1 = 224.1 \text{ N} \quad T_2 = 183 \text{ N}$$

$$T_3 = 336.6 \text{ N} \quad T_4 = 326.8 \text{ N}$$

FBD of B.



$$T_3 \cos 30^\circ = 200 + T_2 \cos 60^\circ$$

$$\frac{T_3 \times \sqrt{3}}{2} = 200 + \frac{T_2}{2}$$

$$\frac{\sqrt{3}}{2} T_3 = 200 + 91.5 = 291.5$$

$$T_3 = 336.6 \text{ N}$$

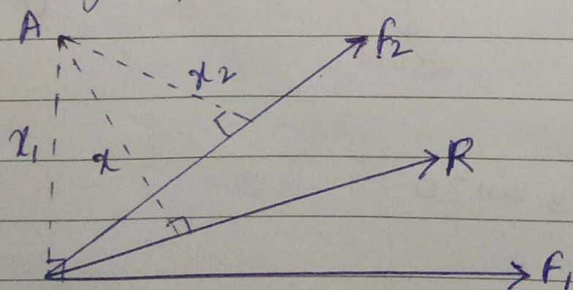
$$T_4 = T_2 \sin 60^\circ + T_3 \sin 30^\circ$$

$$= 183 \times \frac{\sqrt{3}}{2} + \frac{336.6}{2}$$

$$= 158.4 + 168.3 = 326.7 \text{ N}$$

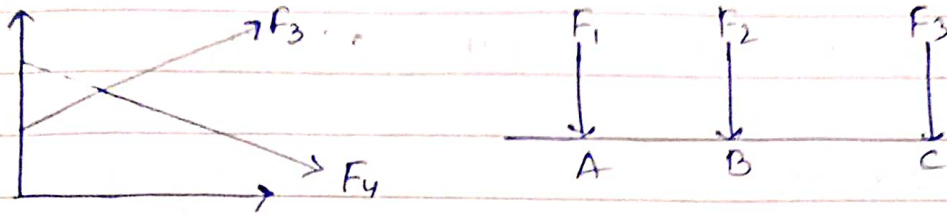
VARIGNON THEOREM:- It states that algebraic sum of moments of coplanar system of forces is equal to the moment of the Resultant force about the same moment centre.

This theorem is used extensively for analysis and stability of structures i.e beams, trusses, frames etc.

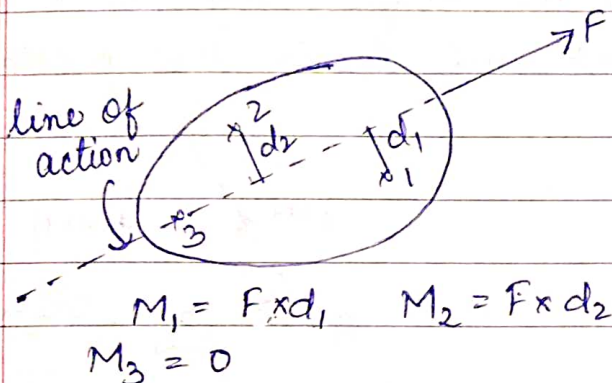


$$\Sigma M_A = R \times x$$

$$\text{or } \Sigma M_A = F_1 x_1 + F_2 x_2 + \dots + F_n x_n$$

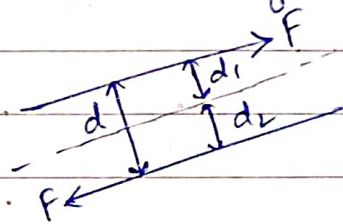


Moment of Force:- It is defined as the measure of rotational effect of force.



Couple:- When two parallel forces equal in magnitude but opposite in direction act along a moment centre it forms a couple.

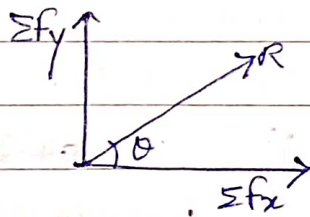
$$M_1 = F(d_1 + d_2) = Fd$$



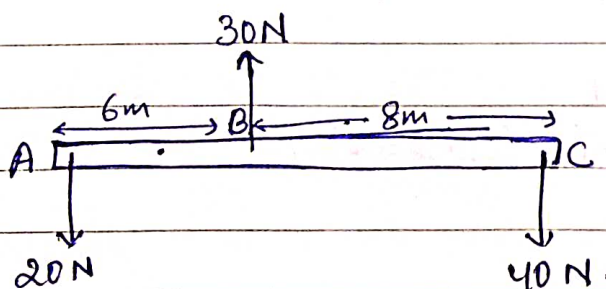
Resultant (R) = $\sqrt{(\sum F_x)^2 + (\sum F_y)^2}$

$\tan \theta = \left| \frac{\sum F_y}{\sum F_x} \right|$

$\theta = \tan^{-1} \left| \frac{\sum F_y}{\sum F_x} \right|$



Que A bar ABC carrying forces 20N at A downward, 30N at B upward and 40N at C downward. Compute the Resultant force and locate its position from point A. AB = 6m, BC = 8m



$$\sum F_x = 0$$

$$\sum F_y = -20 + 30 - 40 = -30 \text{ N} \downarrow$$

$$R = \sqrt{\sum F_x^2 + \sum F_y^2} = 30 \text{ N}$$

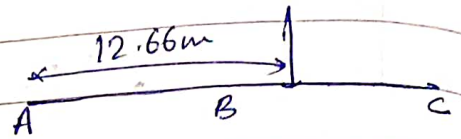
By using Varignon's theorem,

$$\sum M_A = R \times x$$

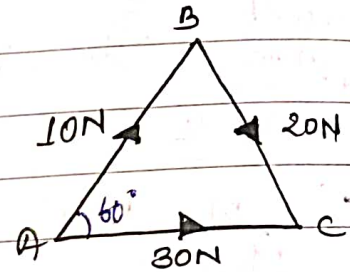
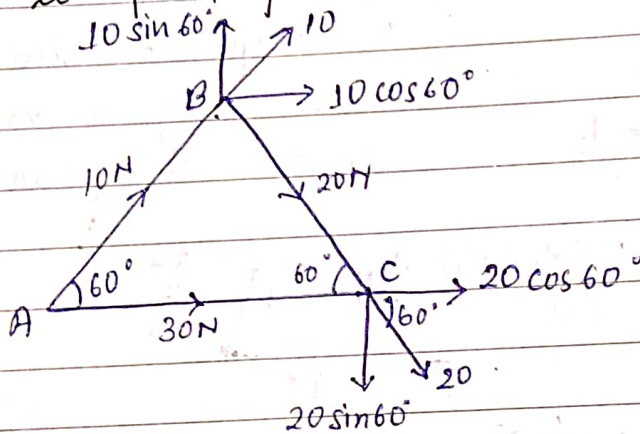
$$\sum M_A = -30 \times 6 + 40 \times 14 = 380 \text{ N-m (clockwise)}$$

$$380 \text{ Nm} = 30 \times x$$

$$x = 12.66 \text{ m}$$



Ques A triangle ABC of 1m side, is subjected to forces 10N, 20N, 30N along AB, BC and AC respectively. Find magnitude, direction of resultant and locate the position of it from point A.



$$\sum F_x = 10 \cos 60^\circ + 30 \text{ N} + 20 \cos 60^\circ = 45 \text{ N}$$

$$\sum F_y = 10 \sin 60^\circ + (-20 \sin 60^\circ) = -8.65 \text{ N}$$

$$R = \sqrt{(\sum F_x)^2 + (\sum F_y)^2} = 48.04 \text{ N}$$

$$\theta = \tan^{-1} \left| \frac{\sum F_y}{\sum F_x} \right| = \tan^{-1} (0.192)$$

$$\theta = \tan^{-1} \left| \frac{8.65}{45} \right|$$

$$\begin{aligned} R &= \sqrt{(45)^2 + (8.65)^2} \\ &= \sqrt{2025 + 74.8225} \\ &= \sqrt{2099.8225} \\ &= 45.823 \end{aligned}$$

$$\theta = \tan^{-1} (0.192)$$

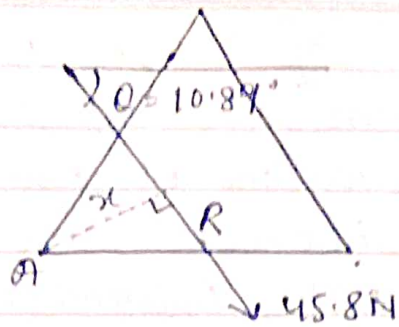
$$\theta = 10.88^\circ$$

$$\sum M_A = 20 \sin 60^\circ \times 1$$

$$= 17.32 \text{ Nm (clockwise)}$$

$$17.32 = 45.8 \times x$$

$$x = 0.377 \text{ m}$$



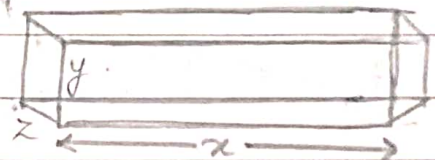
Equilibrium of Non-Concurrent forces.

→ A body is said to be in equilibrium if it has no translational and rotational motion.

$$\Sigma F_x = 0 \quad \therefore \quad \Sigma M_A = 0 \quad \Sigma F_y = 0$$

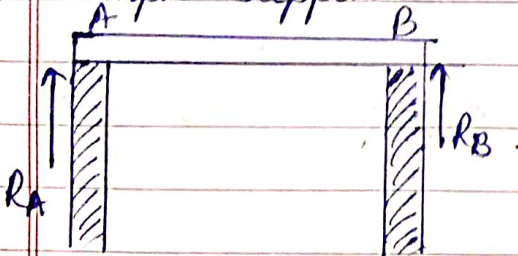
Beam:- Beam is a structural element which has one dimensional length comparatively larger than other two dimension. It is subjected to lateral loads. Hence, there will be support reactions and moments to keep the beam in equilibrium.

$$x \gg y, z$$

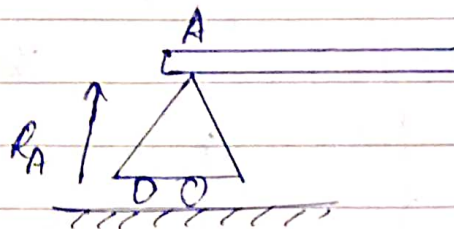


Types of Support

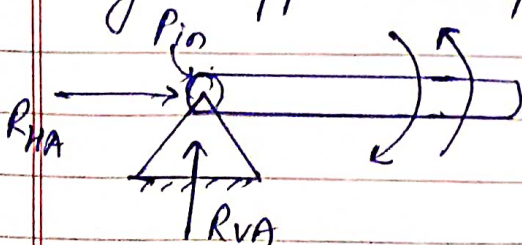
① Simple support



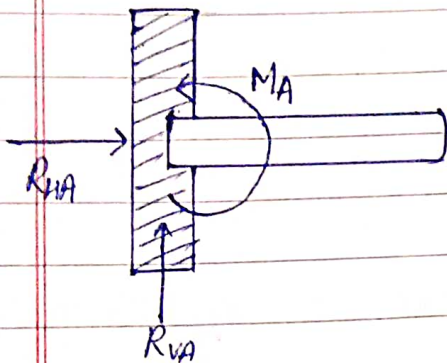
② Roller support



③ Hinge support or pinned support



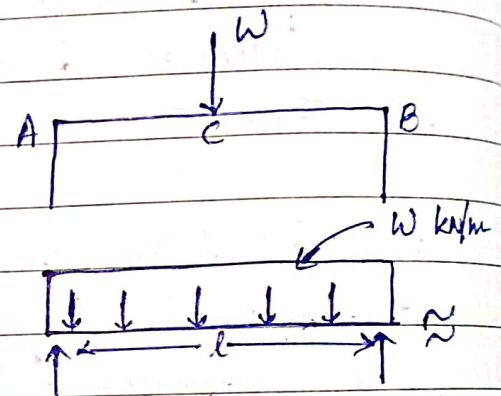
④ Fixed Support



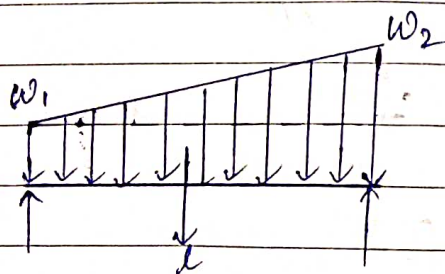
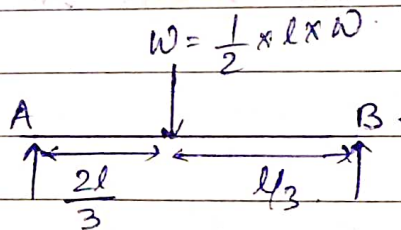
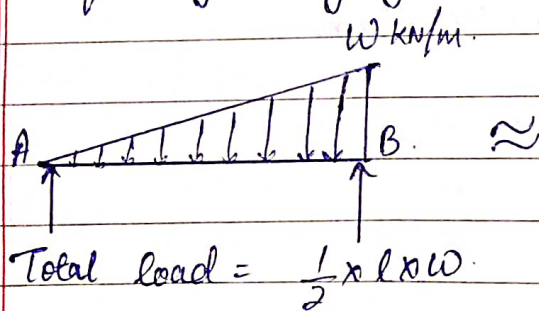
Note:- If all the Reactions of a beam can be solved using the equations of equilibrium, then beam is said to be "Statically Determinate"

Types of Loading

- ① Concentrated or point load
- ② uniformly distributed load
- ③ uniformly varying load



Total load = $W \times l$



Total load = $w_1 \times l + \frac{1}{2} (w_2 - w_1) \times l$